

Substation ESG Report

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS-equivalent, EU Taxonomy reference, TCFD, SFDR-aware, and UN PRI frameworks. Updated annually via the SSI Index scoring engine. Colombia is not an EU member and is not bound by CSRD directly; the framework is applied here for cross-cohort comparability with the OECD-Andean peer set. Adapted to Colombia: CREG independent regulator, UPME planning, Ley 142+143/1994 unbundling, CONPES cyber framework, Just Energy Transition Partnership context.

SUBSTATION

Subestación Cereté

CO_000356 · Córdoba, cordoba

SSI SCORE (R_{MEDIAN})

0.565

● High Risk 9.2th percentile

VOLTAGE CLASS

110 kV

High-voltage transmission

CONFIDENCE INTERVAL

0.533 – 0.599

P5–P95 · CI width 0.088 · high confidence

MARKOV ETTC

38.3 yr

Expected time to criticality

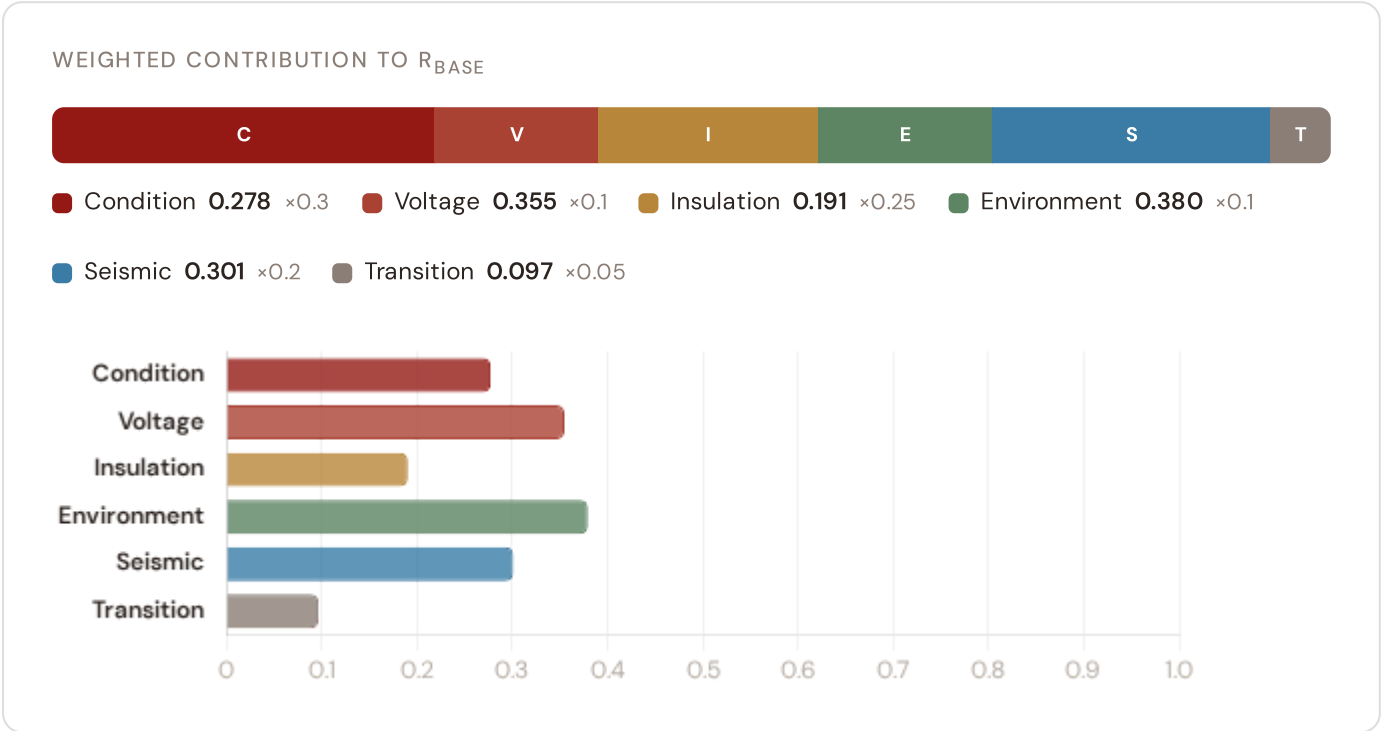
DER PENETRATION

0.11

T1 transition stress score

Component Fingerprint

Six-component weighted decomposition of the SSI composite score — the substation's structural risk profile across the Colombian departamento it serves



Markov Degradation Profile

CIGRE TB 761 condition model with ENSO-modulated transition rates — steady-state probability distribution and expected time to criticality under Colombia's Andean elevation thermal cycling, Caribbean + Pacific coastal salt-aerosol, and ENSO-driven hydro-thermal swing regimes

STEADY-STATE DISTRIBUTION



Markov chain extends CIGRE Technical Brochure 761 (2019) with ENSO-modulated transient states reflecting El Niño thermal-ramp stress and La Niña flood/landslide stress. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition given the Colombian operating environment — dominant drivers are Andean elevation thermal cycling, Pacific Chocó + Caribbean coast C5 ultra-saline corrosion, Cesar mining + Magdalena-medio C4 industrial atmospheres, and reservoir-drawdown thermal-ramp during strong El Niño windows.

RISK SCORE

0.281

Composite Markov risk

ETTC

38.3 yr

Expected time to critical

P(CRITICAL) 20YR

29.7%

Probability of reaching critical

CORROSION

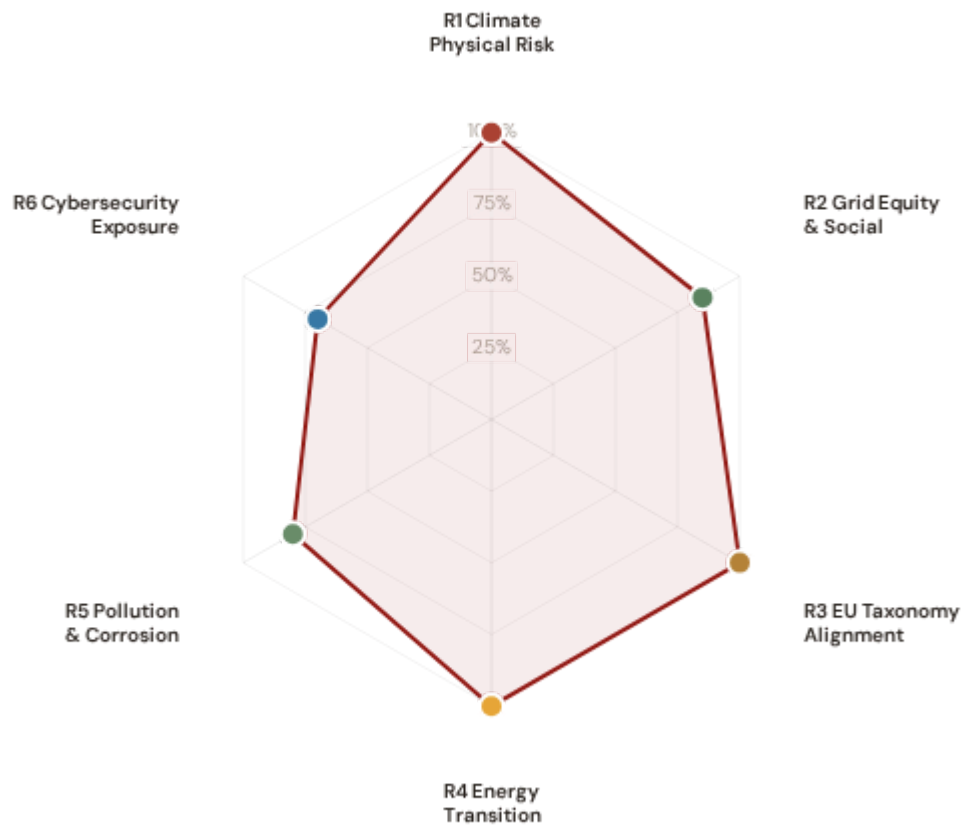
C4

ISO 9223 (C2 Andean inland · up to C5 Pacific Chocó + Caribbean coast)

ESG Radar — Six-Report Overview

Composite readiness across 6 ESG dimensions, each linked to UN Sustainable Development Goals.

Colombia-specific anchors: 12 GW renewable-by-2030 UPME target, coal phase-out 2030-2035 (Cesar + La Guajira JETP context), hydro-dominant matrix with ENSO-amplified thermal swings, ~9,4M IDP + ~3M Venezuelan-refugee social-resilience overlay, and CoLCERT + CSIRT-XM cyber posture under CONPES 3854/2016 + 3995/2020.



● R1 · Climate Physical Risk Assessment READY

SDG 13 Climate Action · SDG 9 · SDG 11

SDG 13 SDG 9 SDG 11

● R2 · Grid Equity & Social Vulnerability READY

SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1

SDG 7 SDG 10 SDG 1

● R3 · EU Taxonomy Alignment READY

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12

SDG 9 SDG 13 SDG 12

● R4 · Energy Transition & DER Stress READY

SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7 SDG 13 SDG 9

● R5 · Pollution & Corrosion READY

SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15

SDG 11 SDG 3 SDG 15

1 Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

READY

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of 0.281 and ETTC of 38.3 years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13

SDG 9

SDG 11

Substation Profile

Insulation Risk (I component)	0.191
Environment Component (E)	0.380
Seismic PGA	0.135g (Zone Caribbean stable)
Markov Risk Score	0.281
ETTC (time to criticality)	38.3 years
P(critical) at 20 years	29.7%
Corrosion Class	C4
Climate Trajectory (R2)	29.7% (20yr Markov)

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
I1 Snow/Ice IRI	—	ESRS E1-9: Financial effects from physical risks	READY
I2 Tree-fall IRI	—	TCFD Strategy (b): Physical risk impact	READY
I3 Heat-wave IRI	—	EU Taxonomy Annex A: Heat stress	READY
I5 Thermal stress	—	ESRS E1-4: Climate mitigation targets	READY
R2 Climate trajectory	29.7% (20yr)	TCFD Strategy (c): Scenario analysis	READY
R6b Seismic PGA	0.135g	EU Taxonomy Annex A: Geophysical hazards	READY
Markov p_critical_10yr	29.7%	TCFD Risk Management: Risk identification	READY
Markov p_critical_20yr	29.7%	ESRS E1-9: Time horizons	READY

Data Sources & Vintage

ERA5 Climate Reanalysis

Copernicus CDS

2024 Weekly

Colombia Seismic Hazard Map

SGC PGA 475-yr · Nazca–South American subduction megathrust + Bucaramanga seismic nest

2023 Multi-year

Volcanic Monitoring

SGC observatories — OVSM Manizales (Ruiz) + OVSP Pasto (Galeras) + OVSPop Popayán (Puracé) networks; 9 active edifices including Nevado del Tolima/Huila/Doña Juana/Cumbal/Sotará/Cerro Bravo

2024 Continuous

Weather + Climate + ENSO

IDEAM (Instituto de Hidrología, Meteorología y Estudios Ambientales)

2024 Daily

Flood + Landslide Mapping

IDEAM + UNGRD (Unidad Nacional para la Gestión del Riesgo de Desastres) — 2010–2011 La Niña + 2017 Mocoa landslide + 2024 La Niña flood anchors

2024 Annual

IEEE C57.91 Thermal Model

IEEE

Standard N/A

CIGRE TB 761 Markov

CIGRE

2019 N/A

CMIP6 SSP2–4.5 Projections

Copernicus CDS

2024 Multi-year

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. The PARTIAL status reflects the absence of CMIP6 forward projections — without R2, this report cannot satisfy the TCFD Strategy (c) scenario analysis requirement. All other inputs are production-grade with institutional provenance.

2

Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

READY

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V_socio of 0.35 and energy poverty rate of 21%. The R3 social modifier (1.027) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7

SDG 10

SDG 1

Substation Profile

V_socio (energy poverty index)	0.35
EP Rate (energy poverty %)	21%
Unemployment Rate	11.5%
GDP per Capita	€ 3,500
R&D % of GDP	0.1%
R3 Social Multiplier	1.027
Elderly Vulnerability	9%
Migration Score	—

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
V_socio Energy poverty index	0.35	ESRS S2-4: Impacts on affected communities	READY
EP_rate Energy poverty rate	21%	SFDR PAI #14: Fossil fuel exposure proxy	READY
R3 Social multiplier	1.027	ESRS S1-6: Workforce characteristics	READY
Elderly Elderly vulnerability	9%	SFDR PAI: Vulnerable populations	READY
Community Catchment population	1,800,000	ESRS S2: Community impact	READY

Data Sources & Vintage

Population & Economics DANE (Departamento Administrativo Nacional de Estadística) — 2018 Census + 2024 mid-year projections (~52.2M)	2024 Annual
Energy Market Data XM Boletín Estadístico + CREG electricity quality indicators	2024 Daily
Internally Displaced Persons + Refugee Unidad de Víctimas IDP register (~9.4M cumulative) + Migración Colombia Venezuelan refugees (~3M)	2024 Quarterly
Cross-Border Interconnect ISA INTERCOLOMBIA + Ecuador (CENACE) — 230 kV Pasto-Quito ~250 MW commercial; Venezuela link suspended 2008; Panama Andean-CA interconnect extension Cerromatoso-Veladero planned	2024 Daily

TECHNICAL VALIDITY

V_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. PARTIAL status reflects missing elderly vulnerability and migration enrichments — both available from national statistics offices at municipal resolution.

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R_median (0.565), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9

SDG 13

SDG 12

Substation Profile

R_median (composite)	0.565
R_base (pre-modifier)	0.398
Modifier Impact	41.5%
C (Condition)	0.278
V (Voltage)	0.355
I (Insulation/Climate)	0.191
E (Environment)	0.380
S (Seismic)	0.301
T (Transition)	0.097

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R_median Composite score	0.565	EU Taxonomy Art. 11: Adaptation screening	READY
6 Comp. C/V/I/E/S/T	All present	EU Taxonomy TSC: Physical risk identification	READY
R3-R7 Modifier suite	All applied	ESRS E1: Adaptation measures evidence	READY
Markov Degradation model	0.281	TCFD: Forward-looking risk assessment	READY
R5 Asymmetric CI	0.088	ESRS: Uncertainty quantification	READY

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
Colombia Seismic Hazard Map SGC PGA 475-yr · Nazca-South American subduction megathrust + Bucaramanga seismic nest	2023 Multi-year
Volcanic Monitoring SGC observatories — OVSM Manizales (Ruiz) + OVSP Pasto (Galeras) + OVSPop Popayán (Puracé) networks; 9 active edifices including Nevado del Tolima/Huila/Doña Juana/Cumbal/Sotará/Cerro Bravo	2024 Continuous
Population & Economics DANE (Departamento Administrativo Nacional de Estadística) — 2018 Census + 2024 mid-year projections (~52.2M)	2024 Annual
Internally Displaced Persons + Refugee Unidad de Víctimas IDP register (~9.4M cumulative) + Migración Colombia Venezuelan refugees (~3M)	2024 Quarterly
CIGRE TB 761 Markov CIGRE	2019 N/A

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

WHY THIS REPORT EXISTS

The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of 0.107. The T1_score (0.483) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7

SDG 13

SDG 9

Substation Profile

T1 Score (transition stress)	0.483
DER Ratio	0.107
DER Variability	0.347
EV Load Ratio	0.021
T Component Weight	0.05

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
T1 Transition stress score	0.483	ESRS E1: Transition plan assessment	READY
DER_ratio Renewable/load ratio	0.107	TCFD Transition: Technology risk	READY
DER_var Intermittency	0.347	EU Green Deal: Grid flexibility	READY
EV_load EV charging ratio	0.021	SDG 7: Clean energy access	READY

Data Sources & Vintage

Energy Market Data

2024 Daily

XM Boletín Estadístico + CREG electricity quality indicators

Renewable Generation Mix

2024 Monthly

XM/UPME (hydro 70% + thermal gas 15% + coal 5% + solar 3% + wind 2% in 2024 normal year; ~50/30/10/3/2 in 2024 El Niño Q1 emergency)

Cross-Border Interconnect

2024 Daily

ISA INTERCOLOMBIA + Ecuador (CENACE) — 230 kV Pasto–Quito ~250 MW commercial; Venezuela link suspended 2008; Panama Andean–CA interconnect extension Cerromatoso–Veladero planned

Decarbonization + Energy Transition

2024 Annual

UPME PERG + MME NDC 2030 (51% emissions reduction vs BAU) + coal phase-out 2030–2035 (Cesar + La Guajira Just Energy Transition Partnership)

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

5 Pollution & Corrosion

READY

ESRS E2 Pollution · ISO 9223 Corrosion Classification · Environmental Impact

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (C4) and E2_local enrichment factor (0.30) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11

SDG 3

SDG 15

Substation Profile

Corrosion Class (ISO 9223)	C4
E2 Local Enrichment	0.30
E Component Score	0.380
Markov Steady-State	27%/26%/22%/15%/9%

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Corrosion ISO 9223 class	C4	ESRS E2: Pollution risk classification	READY
E2_local Environmental sensitivity	0.30	ESRS E2: Local environmental impact	READY
E Environment component	0.380	ISO 9223: Atmospheric corrosion	READY

Data Sources & Vintage

ISO 9223 Corrosion (C2–C5 — C5 Pacific Chocó + Caribbean coast)	2012 N/A
ISO	

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class agency shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO2, NOx, particulate deposition).

6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

PARTIAL

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment,

communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (1.028) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (0.814) identifies single-point-of-failure risk in the grid topology.

SDG 9

SDG 16

Substation Profile

R7 Cyber Modifier	1.028
Graph Degree (topology)	1
BC Percentile (centrality)	0.814
Is Bridge Node	No
Cluster Coefficient	0.2570

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R7 Cyber modifier	1.028	NIS2: Cybersecurity risk management	READY
BC Betweenness centrality	0.814	ESRS G1: Governance & topology risk	READY
Degree Graph degree	1	Grid resilience: Connectivity	READY

Data Sources & Vintage

Cybersecurity Posture

2024 Annual

CoICERT (MinTIC+MinDef, 2011) + CSIRT-XM sectoral electrical (XM, 2019) + CCOCI (Cyber Defence Command, MinDef) + CONPES 3854/2016 + 3995/2020

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

C Audit Trail & Methodology Note

Traceability for CSRD Article 29a-equivalent limited assurance — Colombia non-EU disclosure reference

ITEM	VALUE
Scoring Engine	SSI Index v4.0.2 — Gaussian copula Monte Carlo (10,000 iterations)
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R3: 1.020, R4: 1.000, R6 total: 0.350, R7: 1.028, R6 seismic: 0.050, R6 volcanic: 0.000, R6 drought: 0.050, R6 armed_conflict: 0.150, R6 flood: 0.100, R3 C_mult: 1.027, R4 F_topo: 1.000, R6 restoration: 1.105, R7 cyber: 1.028
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Subestación Cereté (CO_000356)
Data Assurance Level	All inputs from institutional open-source with citable vintage